

Reliability-Aware Clinical AI: A Cross-Modality Contract for Cardiovascular Decision Support

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Abstract—Clinical machine learning is usually reported as one accuracy figure, and the software around the model then treats every prediction as equally trustworthy. We argue a deployed model must also state whether each prediction is reliable enough to act on, and we make that statement a first-class output. We define a reliability contract: a modality-independent mapping from a prediction, its uncertainty, its input quality and its validity conditions onto five ordered decision states, actionable, caution, deferred, withheld and unavailable, so a caller applies one rule across every modality. We instantiate the contract on four separate cardiovascular cohorts, chest radiographs (MIMIC-CXR), 12-lead electrocardiograms (PTB-XL), echocardiogram video (EchoNet-Dynamic with CAMUS) and emergency-department triage records (MIMIC-IV-ED), and test every mechanism against a control that keeps the procedure and removes the signal. Three findings recur. Acquisition metadata drives a 6.68 point accuracy gap that group-conditional deferral closes to -0.62, where uniform deferral of the same budget does not. A conformal guarantee that holds over the population fails inside 9 of 23 sex and age subgroups, which group-conditional calibration repairs in 22 of 23. Admitting a feature only once it existed at the decision time moves screening AUROC from 0.8763 to 0.9560 and exposes a label-circularity path we would otherwise have reported as accuracy. Under one common metric, the unsafe answer rate, abstention lowers unsafe answers in every component where coverage is measurable. Four ablations of our own design choices return three negative results.

Index Terms—abstention, clinical decision support, conformal prediction, covariate shift, data leakage, model reliability, selective prediction, subgroup validity, trustworthy AI

I. INTRODUCTION

A patient who arrives at an emergency department with chest pain generates four very different kinds of data within a day. A triage nurse records vital signs and a free-text complaint. An electrocardiogram is recorded within ten minutes [1], because a blocked artery has to be found before any blood test can come back. A chest radiograph is taken early, mainly to rule out causes that are not cardiac at all [2]. Blood is drawn and repeated, and an ultrasound scan of the beating heart is booked. Each has become a machine-learning problem in its own right, and in each case the published result is normally one accuracy figure on one dataset.

One figure is a poor description of a model that is about to sit behind an API. It says nothing about the inputs on which the model is weakest, and in each of our four tasks those inputs are not rare corner cases but the sickest patients, an identifiable subgroup, the most severe grades, or a feature that did not exist when the decision was made. Table I names all four with the mechanism each one motivated. One is worth stating here, because it is why this paper exists: an audit of an earlier version of our own triage

component found it reading its label out of a comorbidity column, scoring well on information the clinician would not have had, and we rebuilt it from the raw tables.

These four problems have separate literatures, but the engineering response is the same each time: the model has to expose when it should not be believed, in a form the calling code can branch on. We therefore treat reliability as the object of study rather than a property of any one network. Three questions organise the work. *RQ1*, can reliability-aware mechanisms reduce clinically important failure modes without materially reducing predictive performance? *RQ2*, can mechanisms as different as a group threshold, a conformal zone, a prediction interval and a disclosure horizon be represented by one common contract? *RQ3*, does reliability-aware abstention produce safer decisions than answering everything, or than deferring uniformly? The contributions are:

- a *reliability contract*: five ordered decision states with a modality-independent rule for assigning them, so heterogeneous uncertainty vocabularies reduce to one field a caller can branch on without knowing how any component works;
- a cross-modality evaluation of that contract on four separate cardiovascular cohorts under one common measure, the unsafe answer rate, reported beside coverage so the cost of abstaining is priced rather than hidden;
- controlled reliability experiments, one measured failure mode per modality, each with a control arm that keeps the procedure and removes the signal: acquisition shift, subgroup validity of a conformal guarantee, silent input corruption, open-set inputs, tail shrinkage under imbalance and temporal leakage;
- four ablations of our own design choices, three of them negative results we report rather than bury, including a squeeze-and-excitation block that costs accuracy.

II. RELATED WORK

Each modality has an established model family: convolutional classification over MIMIC-CXR [3] with a ConvNeXt backbone [4] and a BioBART decoder [5]; one-dimensional residual networks over PTB-XL [6]; video regression on EchoNet-Dynamic [7] with an R(2+1)D backbone [8], with CAMUS [9] as a smaller cohort richer in severe cases; and gradient-boosted trees [10], [11] with post-hoc attribution [12] over emergency-department tables [13]. Saliency is usually Grad-CAM [14].

The reliability side has its own literature, which we use rather than reinvent. Subgroup performance gaps are well documented [15]; equal opportunity [16] is the usual metric, and fitting one threshold per group is the post-processing method proposed alongside it, so on this axis we claim the measurement and not the technique. The common alternative is to train the offending factor out of the

representation, for instance with label-conditional gradient reversal [17]. A separate line lets the model abstain, from the reject option [18] to conformal prediction, which turns a score into a decision carrying a finite-sample bound [19], [20], usually after calibration [21]. Closest to our deferral result is [22], which shows abstention can *widen* group disparities; that is what our uniform-deferral control does, and the gap to the group-conditional version is our result. Ordinal targets use rank-consistent heads [23], long tails deferred re-weighting [24], and leakage was formalised for data mining generally [25].

What we did not find was these ideas used together, across more than one modality, with every mechanism checked against a null arm and the guarantee reported other than marginally.

III. THE RELIABILITY CONTRACT

A component that cannot say when to disbelieve it forces the caller to guess, and every caller guesses differently. The contract makes that judgement an output. Each component keeps its own weights, its own uncertainty statistic and its own frozen decision rule; what it must additionally emit is one field drawn from five ordered states:

- *actionable*, the component stands behind the result and measured reliability for this kind of input is the reliability we report;
- *caution*, the answer stands, but a validity condition of that measurement does not hold here, so reliability is lower than headline;
- *deferred*, the component declines to commit and the case is referred;
- *withheld*, output was suppressed because a quality or verification gate failed, so no probability is released at all;
- *unavailable*, the component could not run on this input.

The states are ordered, most to least usable, and that ordering is the whole interface: a caller applies one rule, do not act on a result that is not actionable, without knowing what a projection, a conformal zone or a disclosure horizon is. Assignment is a precedence cascade over four signals a component already computes: whether it ran, r ; whether the input passed its quality and verification gates, q ; an uncertainty statistic u against a threshold τ fitted on validation and frozen; and whether the validity conditions of the reported reliability hold for this input, v . Numbering the states 0 to 4 in the order above, component m returns

$$a_m = \max(0, 1[-v_m], 2[-q_m], 3[-u_m < \tau_m], 4[-r_m]), \quad (1)$$

so the least usable applicable state wins and no component can be made to look safer by an ordering accident. Over the set $M(x)$ of components that saw patient x , the assessment takes the same maximum,

$$a(x) = \max_{m \in M(x)} a_m(x), \quad (2)$$

which aggregates and deliberately does not fuse. Substituting a component changes only what fills u , q and v ; Section IV gives four such substitutions. The states are returned as a normal response, never an error status, since turning a safety mechanism into an error pushes callers into retry loops around it.

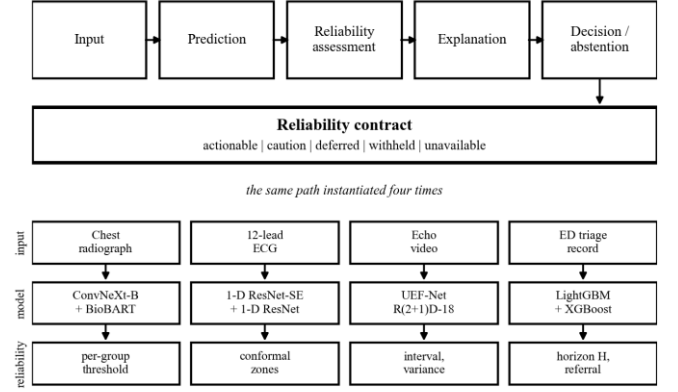


Fig. 1. The framework. One path runs left to right for every modality; the four cardiovascular components are instantiations of it, differing only in what fills the uncertainty, quality and validity slots.

That the endpoint aggregates rather than fuses is a limitation we measured, not a preference. Of the six cohort pairs only radiograph and triage are linkable, both deriving from MIMIC-IV and sharing 19,979 patients, 81.6 % of the radiograph cohort; the other five share zero by construction, since PTB-XL, EchoNet-Dynamic and CAMUS come from different hospitals, countries and decades. No patient in any cohort carries all four studies, so this is a cross-modality framework evaluated on four separate cohorts, not patient-level multimodal fusion, and we train no fusion model and report no joint accuracy.

IV. INSTANTIATING THE CONTRACT IN FOUR MODALITIES

Each subsection below fills the same four slots for one modality: the quality gate q , the uncertainty statistic u and its frozen threshold, the validity condition v whose failure downgrades a result to caution, and the control arm the mechanism is tested against. The models themselves are standard; the reliability machinery around them is the contribution.

A. Covariate Shift from Acquisition Metadata

A ConvNeXt-Base backbone [4] with a two-layer head produces eight sigmoid outputs from a 384 by 384 image, standardised per image rather than with ImageNet statistics, which are wrong for a single-channel radiograph. Grad-CAM [14] is taken at the last feature block, and a BioBART decoder [5] drafts report text we do not evaluate here. The interesting part is not the network. Every radiograph carries a metadata field g in $\{AP, PA\}$ recording how it was taken; the two groups are not the same distribution, so one operating point for both is a modelling error. Following [16], we fit one per group on validation data only,

$$\tau_g = \arg \max_{\tau} F1(y_g, [p_g \geq \tau]), \quad (3)$$

and apply the threshold belonging to the image's own group at inference. Ranking cannot change under (1), since AUROC is computed over the whole ordering and cutting each group at a different point reorders nothing, verified numerically to twelve decimal places. On top of it a selective rule refers the case when the prediction sits too close to the operating point,

$$m = |p - \tau_g|, \text{ answer if } m \geq q_g, \text{ otherwise refer}, \quad (4)$$

with $q(\text{AP})$ and $q(\text{PA})$ fitted on validation under one shared coverage budget, choosing the pair that minimises the absolute accuracy difference between groups, and then frozen. At the deployed 85 % coverage target they are 0.2247 and 0.0029, so the system is far more reluctant to commit on the weaker group. An image with no projection field falls back to the global threshold.

B. A Guarantee, and Two Ways It Silently Stops Holding

A quality gate checking shape, duration, dead leads, units, amplitude, noise and rhythm runs before the classifier, so a rejected recording never produces a probability. Accepted recordings are band-pass filtered from 0.5 to 40 Hz with a 50 Hz notch, resampled to 500 Hz and normalised per lead, then classified by a one-dimensional residual network with squeeze-and-excitation and attention pooling, followed by per-class temperature scaling [21]. A triage layer turns each calibrated probability into one of three decisions under a training-conditional, or PAC, conformal bound [19]. For a class with n calibration positives, miss-rate budget α and confidence δ , the order statistic is

$$m^* = \max \{ k \leq n : F_{\text{Beta}}(\alpha; k, n - k + 1) \geq 1 - \delta \}, (5)$$

since the coverage of the k -th order statistic follows $\text{Beta}(k, n - k + 1)$. The rule-out threshold is that order statistic among the positive calibration scores, the rule-in threshold its mirror over the negatives at a false-alarm budget β , and anything between them is referred. A class with too few calibration positives is reported unattainable rather than approximated. Two models are served side by side, each with its own calibrator and thresholds, and a class is ruled out only when both rule it out, so the merged miss rate is bounded by the tighter single-model bound at the cost of more referrals.

A bound like this is conditional on assumptions the input can break without looking broken, so two checks withdraw the guarantee while leaving the prediction in place. The first flags a swapped pair of limb electrodes, an exact linear map of the standard lead definitions, from the polarity of one lead and the inversion of another; the second asks whether the rhythm is inside the label space at all, from a beat-interval irregularity score thresholded on validation at a 5 % false-positive budget. Either way the probabilities are still returned and only the bounded-miss-rate claim is withdrawn, which is the *caution* state of (1) rather than *withheld*.

C. Ordinal Targets, Noisy Labels and a Long Tail

An R(2+1)D-18 backbone [8] pretrained on Kinetics-400 takes 32-frame clips of 112 by 112 pixels in two channels, grey level and a temporal difference, and four heads: regression, ordinal cumulative, auxiliary class and log-variance. The boundaries at 30, 40 and 55 are clinical conventions, and the label they cut is itself noisy, since two readers typically disagree by about 4 points. Treating it as exact throws information away, so the ordinal targets are soft,

$$s_k = 1 - \Phi((t_k - e) / \sigma), (6)$$

where e is the recorded value, t_k the k -th boundary, $\sigma = 4$ and Φ the standard normal distribution function, so s_k is the probability that the true value lies above t_k given a noisy measurement. Rank consistency is structural instead

of repaired afterwards, unlike [23]: one severity score is compared against cut-points that increase by construction, because each gap is a softplus,

$$c_l = a, \quad c_k = a + \sum_{j < k} \text{softplus}(g_j), \quad z_k = f(x) - c_k, (7)$$

so the cumulative probabilities can never cross. Training uses a class-balanced sampler with deferred re-weighting [24] from epoch 15 and an exponential moving average of the weights. The second cohort is intensity-matched before being mixed in, since a balanced sampler over-draws from it and would let the network use scanner brightness as a shortcut for severity. A regressor on a skewed target also shrinks predictions towards the mean, pushing the severe tail over the boundary at 30, so an expansion is fitted on validation and applied without touching the weights,

$$\hat{e}' = \bar{y} + \kappa(\hat{e} - \bar{x}), \quad \kappa = \text{sd}(y) / \text{sd}(\hat{e}), \text{ clipped to } [1.0, 1.7], (8)$$

after which the boundaries are re-optimised on validation lexicographically: worst-class recall, then balanced accuracy, then macro-F1. At inference a study is sampled into ten clips and averaged over three seeds, and the interval is split-conformal, widened by the learned aleatoric term and by disagreement between clips.

D. An Information-Availability Contract

The rebuild of this component started from the leak described in Section I, so every feature now declares an availability time $a(f)$ relative to arrival. At a disclosure horizon H the admitted feature set is

$$F_H = \{f : a(f) \leq H\}, (9)$$

and values are clipped to those recorded within H hours. The same cohort, split and code are featurised at $H = 0, 6$ and 24 hours, making accuracy against time a reported axis rather than an unstated assumption. Detection is a mean blend of a LightGBM and an XGBoost ensemble [10], [11]; subtyping uses one four-class model instead of a cascade, since a cascade compounds error, a patient the screen misses never being recoverable later. The operating point is a stated optimisation rather than a hand-tuned multiplier,

$$w^* = \arg \max_w \text{macroF1}(\arg \max_k w_k p_k) \text{ s.t. } \min_k \text{recall}_k(w) \geq \rho, (10)$$

with the recall floor $\rho = 0.75$, solved on validation over bootstrap resamples and then frozen. A case whose top-two margin falls below the $(1 - C)$ quantile of the validation margins is referred to a clinician instead of being subtyped. Results are reported on the intended-use population, meaning visits with a cardiac complaint or an early electrocardiogram order; both are observable at triage, so this is selection, not leakage. A separate head asks which wall of the heart the infarct involves, on a label rebuilt from ICD-9 and ICD-10 diagnosis codes.

V. EXPERIMENTAL SETUP

Four public cohorts are used, every split patient-disjoint and quoted as train / validation / test. C1: MIMIC-CXR-JPG [3], 36,362 / 4,474 / 4,722 images, the test fold holding 2,891 bedside and 1,831 standing films, positive class enriched to 50.4 %. C2: the official PTB-XL [6] folds, 13,801 / 1,709 / 1,711 recordings, keeping only codes at full likelihood. C3: EchoNet-Dynamic [7], 7,465 / 1,288 / 1,277

studies, severity classes at 5.9 / 7.2 / 18.0 / 68.9 %, plus 1,000 CAMUS [9] clips in training only. C4: MIMIC-IV-ED [13], 142,111 / 30,453 / 30,452 stays grouped by patient, 2.65 % positive. Each test split was evaluated once, and every decision rule in Section IV was fitted on validation and frozen before that split was opened. Two components were trained on an NVIDIA L4 and two on an RTX 4060 laptop GPU. Metrics follow the task: AUROC, sensitivity, specificity and true-positive-rate disparity between acquisition groups [16] for C1; per-class recall and NPV with the empirical miss rate against the promised bound for C2; mean absolute error, R^2 and minimum per-class recall for C3; AUROC, NPV and minimum per-class recall for C4. Coverage, the fraction answered rather than deferred, is quoted beside every selective number, since accuracy on the answered subset is not the accuracy of the system.

Differences are tested, not eyeballed. Two rules scoring the same items are compared with McNemar's test [26] in mid- p form with Holm correction [27] inside each family; gaps and aggregate metrics use a paired bootstrap of 10,000 resamples on identical indices. Subgroup miss rates use exact binomial tests with Wilson intervals and Holm correction across all 23 cells; triage intervals are cluster bootstraps resampled by patient.

VI. RESULTS

Results come in four parts: headline detection accuracy per component, what each abstention rule buys against its control arm, four ablations of our own design choices, and a check that the deployed service reproduces the offline numbers.

A. Detection Performance

Detection accuracy is a precondition, not the contribution, so we state it once. On its own test fold each component reaches: C1 cardiomegaly AUROC 0.9189 at 92.3 % sensitivity, $n = 4,722$; C2 macro accuracy 0.864 and recall 0.810, every class above 0.75, $n = 1,711$; C3 mean absolute error 3.979 ejection-fraction points and worst-class recall 0.723, $n = 1,277$; C4 screening AUROC 0.9560 at 99.41 % negative predictive value and subtyping macro-F1 0.7448, $n = 30,452$. Four tasks on four cohorts: the rows are not comparable with each other, and none is compared with a published benchmark, for the split reasons in Section VII.

B. One Contract, Four Instantiations

Table I answers RQ2 in one view: four failure modes with nothing in common, four mechanisms with nothing in common, and one contract state emitted by each. Every mechanism is stated against a control, an arm that keeps the procedure and removes the signal, so the improvement column is a difference and not a level.

TABLE I

THE SAME CONTRACT INSTANTIATED FOUR TIMES. THE ACTION IS THE STATE THE CONTRACT ASSIGNS WHEN THE MECHANISM FIRES; THE IMPROVEMENT IS MEASURED AGAINST THE CONTROL ARM, NEVER AGAINST A PUBLISHED BENCHMARK.

CO MP.	FAILURE MODE	MECHANISM AND ACTION	CONTROL	IMPROVEMENT
C1	Acquisition shift, AP vs PA	Per-group threshold; defer inside the margin (defer, caution)	Uniform deferral, same budget	Gap 6.68 to -0.62 points

C2	Marginal bound invalid in subgroups	Group-conditional conformal zones; two-model consensus (withhold, refer)	Marginal conformal	Valid in 9 of 23 cells to 22 of 23
C3	Tail shrinkage at 11:1 imbalance	Ordinal head, variance expansion, boundary deferral (defer)	Full-coverage prediction	Severe recall 0.590 to 0.687
C4	Temporal leakage	Disclosure horizon H; constrained decision layer (refer)	Horizon sweep H = 0, 6, 24 h	AUROC 0.8763 to 0.9560

C. The Cost of Abstaining

Accuracy on the answered subset flatters any system that abstains, so we report one measure that means the same thing in every modality. The *unsafe answer rate* is the probability that the system both answers and is wrong, $U = c(1 - A)$ for coverage c and accuracy A among answered cases; a component that never abstains has U equal to its error rate, and abstaining can only lower U by lowering c . Reported with coverage it prices abstention instead of hiding it. C1 moves from $U = 16.81$ % at full coverage to 9.64 % at 80.6 % coverage under group-conditional deferral; uniform deferral of the same budget reaches a marginally lower 8.89 % but leaves the acquisition gap intact, which is the trade the contract is meant to make visible. C3 moves from 27.02 % to 20.36 % at 88.4 % coverage, and the 148 deferred studies are genuinely the hard ones, scoring 42.6 % against 73.0 % overall. C4 moves from 21.89 % to 6.68 %, at the price of deferring a third of subtyping decisions. Answering RQ3, abstention lowered the unsafe answer rate against the answer-everything arm in all three components where coverage is measurable, and in C1 the group-conditional arm was the only one that also removed the failure mode. Fig. 2 shows one result per component.

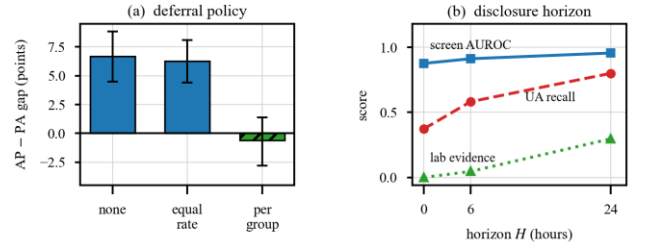


Fig. 2. (a) Accuracy gap between acquisition groups under three deferral policies, with 95 % bootstrap intervals. (b) Screen AUROC, unstable-angina recall and the share of attribution carried by laboratory features, against the disclosure horizon.

Acquisition shift. C1 scores AUROC 0.8224 on bedside images against 0.8864 on standing ones, a gap of 0.0639 [0.0491, 0.0790] in the same direction for all eight labels. Fitting the operating point per group cut the reported true-positive-rate disparity by 73.3 % with an AUROC spread of exactly zero, at no significant accuracy cost (+0.02 points, $[-0.26, +0.31]$, McNemar mid- p 0.885). We also reimplemented the representation-side alternative [17] on our own data, backbone and split: it reached complete invariance, projection-detection AUC 0.5000, and still made the disparity 25.4 % worse at a cost of 0.0789 AUROC. Deferral behaves the same way (Fig. 2a). Deferring the same fraction of both groups leaves the 6.68-point gap at

6.28, the behaviour [22] describes; deferring per group at matched coverage closes it to -0.62 $[-2.78, 1.37]$, a difference of 5.83 points with paired bootstrap $p = 0.0004$, at 85.8 % coverage overall.

Conditional validity. Fitted marginally, the conformal bound held in only 14 of 23 class-by-subgroup cells, and two violations survive Holm correction: one class at a miss rate of 0.333 against a promised 0.10 under age 50, another at 0.330 against 0.20 at age 70 and over, with adjusted p of 5.1×10^{-6} and 0.029. Refitting one threshold per subgroup [20] restored the bound in 22 of 23 cells, at the cost that every cell now needs its own positives: one with 42 cannot support a finite threshold and is reported as unattainable.

Silent corruption and open-set inputs. Simulating each of the three limb-electrode swaps on 200 test recordings, the corrupted signal passes the quality gate in 197 to 198 of them, because it is a clean recording wired wrongly; up to 87 % of diagnoses change and 7 guarantees are voided. The physiology check catches 65.5 % and 60.5 % of two swaps at 4.5 % false positives, and 4.0 % of the third. Separately, 114 recordings carry a rhythm the label space cannot express and 113 received a bounded rule-out for a disease the model has no output unit for; the irregularity gate withholds the claim on 48.9 % of them.

Shrinkage under imbalance. On identical weights, the expansion in (6) lifted recall on the rarest class from 0.590 to 0.687, and seed averaging carried the worst class to 0.723. Selective prediction, which helped C1, failed here: at 88.4 % coverage worst-class recall fell to 0.706 while overall accuracy rose to 0.770. The uncertainty signal is fine, since accuracy on deferred studies is 0.426 against 0.770 on answered ones; the problem is geometric, in that one class occupies a 10-point interior band and abstention removes its members first.

Temporal leakage. One comorbidity column equals 1 for every positive stay and reaches AUROC 0.9200 alone; adding it back to an otherwise safe feature set moves the screen from AUROC 0.9665 to 0.9889, which is how an apparently excellent result gets manufactured. A random split puts 5,804 patients on both sides and contaminates 7,627 test rows; the patient-grouped split shares none. Under the availability contract, performance becomes a function of time (Fig. 2b): screen AUROC 0.8763, 0.9121 and 0.9560, and recall on the hardest subtype 37.3 %, 58.2 % and 80.0 % at $H = 0, 6$ and 24 . That subtype is defined by a normal blood test, so it cannot be separated from its neighbour until the test returns. At $H = 0$ the laboratory channel carries exactly 0.0 % of the attribution mass, rising to 4.6 % and 29.6 %; a leaking pipeline cannot produce that pattern.

D. Turning the Same Standard on Our Own Designs

The mechanisms above were tested against controls. So were four of our own design decisions, and three came back negative. The sharpest is C2's architecture, which adds three things to a plain one-dimensional residual network. At three seeds each, compared by paired bootstrap on the untouched fold, they do not earn their 566 k parameters: the stem and attention pooling change nothing ($p = 0.741$), and squeeze-and-excitation on top of them costs 0.0042 macro-AUROC ($p = 0.0040$). Almost the whole loss sits on one class, $+0.0147$ AUROC without it, and there is a mechanism rather than a coincidence: that diagnosis is read from QRS

amplitude, and squeeze-and-excitation recalibrates channels by learned importance, an operation on relative amplitude across leads.

C4's infarct-wall head is the second. With every feature it reaches AUROC 0.9074 and no per-class metric below 0.7551 on 104 test cases; removing three features parsed from the recording device's own printed interpretation costs 0.133 AUROC and drops the weakest metric to 0.6038, and those three alone reach AUROC 0.841. This is not temporal leakage, since they exist at triage, but the person who assigned the diagnosis code read the same printout. Feature and label therefore share a source: the label is partly defined by an input the model is given, a circularity no timestamp check can detect, and both numbers belong in any report of this head. Widening it beyond two territories was measured too: a third class is recalled in 1 case of 12 and drags the other two down. The remaining two ablations are quieter. C3's backbone was tested against the un-factorised alternative at three matched seeds and is worth keeping, though only on the classification metrics. Text generation exists in two components but is not evaluated here, and neither it nor the wall head is served.

One check of the same kind closes the loop. Research and serving code drift apart, so beside the service's 132 automated tests the radiograph endpoint was scored on 200 stratified real studies posted through the live HTTP path. Served accuracy was 0.790 [0.728, 0.841], sensitivity 0.880 [0.802, 0.930] and specificity 0.700 [0.604, 0.781], all consistent with the offline figures at 14.0 % deferred, and accuracy was 0.766 on bedside images against 0.833 on standing ones. The covariate shift reappears on real inputs through the deployed path, and the horizon contract is enforced per patient: for a single record the laboratory channel reads 0.000 % at triage.

VII. DISCUSSION, THREATS TO VALIDITY AND LIMITATIONS

Answering RQ1 and RQ3 together: in these experiments the effective intervention sat in the decision layer rather than the representation. Three attempts to close the acquisition gap by changing the model failed against a null arm, while a threshold and a deferral budget conditioned on the same variable worked, and the unsafe answer rate fell in every component where coverage is measurable. We do not conclude that clinical reliability is generally post-processing. We conclude something narrower: on these four tasks, several reliability problems were addressed effectively by decision-layer controls that are cheap to fit, auditable and changeable without retraining, and conditioning each control on the variable that actually causes the failure mattered more than the strength of the control. RQ2 is answered by construction and then tested by use: four mechanisms with nothing in common reduced to one five-state field without any of them losing information, since the component-native payload is returned unchanged alongside it.

Threats to validity. All four cohorts are retrospective and public, so distribution, labelling convention and case mix reflect the institutions that released them; MIMIC-CXR and MIMIC-IV-ED come from one US hospital, PTB-XL from a German cohort of the 1990s, EchoNet-Dynamic from one US centre and CAMUS from one French one, and no result here transfers to another site without being re-measured.

Every reliability threshold is fitted on validation and frozen, so it inherits that cohort's case mix and would need refitting elsewhere; the coverage targets and the recall floor $\rho = 0.75$ are chosen by us, not derived from a clinical standard. There is no external validation, no prospective evaluation, no clinician-in-the-loop study and no patient outcome measured, so we can report that a decision was withheld but not whether withholding it helped anyone. The absence of fully paired multimodal data means the aggregation rule of Section III is tested per component and not end to end.

Limitations. The C1 split is custom, its positive class enriched to 50.4 %, and 98.3 % of its test images fall inside the official MIMIC-CXR training split; we therefore treat it as an internal operating point, make no comparison with published MIMIC-CXR benchmarks anywhere in this paper, and put a strict patient-level holdout first in further work. C2 keeps only codes at full likelihood and drops 21 % of PTB-XL, so the same restriction applies. Training variance is measured for C2 and C3 only. Subgroup coverage is partial: C1 has the acquisition field, C2 sex and three age bands, C3 no demographic fields, C4 has them but no breakdown yet. The electrode audit rests on 200 recordings, and regular out-of-scope rhythms remain silent failures. The infarct-wall head has 104 test cases. C3's worst-class recall improved but did not reach the 0.75 target we set, on test or on validation. Grad-CAM is used as a sanity check, not as evidence of localisation: its repeatability on chest radiographs has been measured at a structural similarity of 0.12 [28], so we make no explainability claim beyond that. This is a retrospective research prototype, not a clinically validated system and not a medical device.

VIII. CONCLUSION

A deployed clinical model should not only produce a prediction; it should state whether that prediction is reliable enough to act on. We made that statement a first-class output through a reliability contract, five ordered states assigned by a precedence rule over a component's own quality, uncertainty and validity signals, and instantiated it in four cardiovascular modalities that share no patients and no features. Conditioning abstention on the variable that actually causes the failure beat changing the model in every comparison we ran, and lowered the unsafe answer rate wherever coverage could be measured. The same standard applied to our own design choices returned three negative results out of four. What the contract buys is that a caller need not know any of this: it applies one rule everywhere. Next are external validation, a strict patient-level holdout for C1, multi-seed training for the two components that lack it, and a paired study on the 19,979 patients our radiograph and triage cohorts share.

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